

Course 2 overview



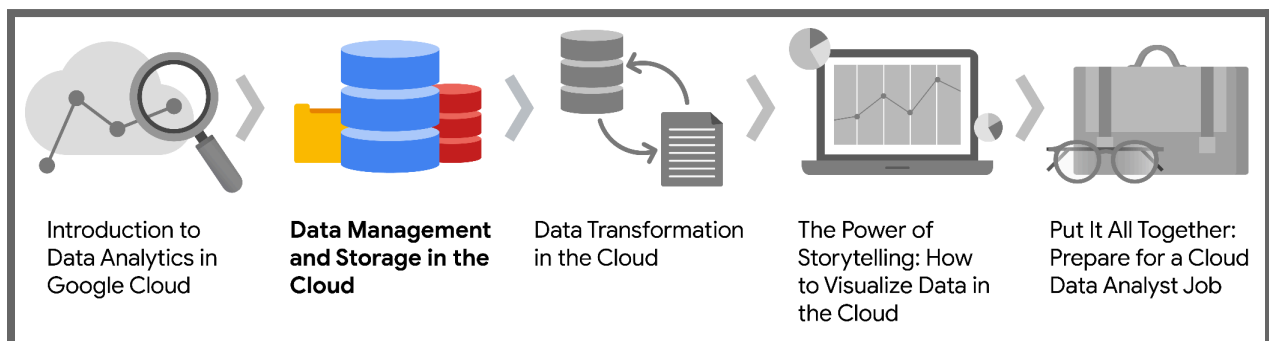
to Course 2

Hello and welcome to **Data Management and Storage in the Cloud**, the second course in the Google Cloud Data Analytics Certificate.

In this course, you'll develop a greater understanding of how data is structured and organized. First, you'll explore how data is stored and moves across systems. Then, you'll learn about different types of data, their storage structures, and how data can be extracted. Next, you'll explore the key components of data governance, and learn about normalized and star schemas, data catalogs, and data lakehouse architecture. You'll also get a chance to work directly with Google's cloud data storage and management tools, including BigQuery and Dataplex. Finally, you'll learn how to interact with data tables, and use optimization techniques to improve queries.

Certificate program progress

The Google Cloud Data Analytics Certificate program has five courses. **Data Management and Storage in the Cloud** is the second course.



1. Introduction to Data Analytics in Google Cloud — In this course, you'll explore the foundations of cloud data analysis practices. Then, you'll learn more about the roles and responsibilities of a cloud data analyst, including data acquisition, storage, processing, and visualization.
2. **Data Management and Storage in the Cloud** (current course) — In this course, you'll learn more about how data is structured and organized. You'll also gain hands-on experience with data lakehouse architecture and cloud components like BigQuery, Google Cloud Storage, and DataProc to learn how to efficiently store, analyze, and process large datasets.
3. Data Transformation in the Cloud — In this course, you'll use data discovery and profiling methods to examine data quality and validity. You'll also use data mapping to establish relationships between different data sources, and explore transformation using various methods. Then, you'll apply data governance and security best practices to keep PII data safe, and learn how to implement a data transformation strategy for an organization.
4. The Power of Storytelling: How to Visualize Data in the Cloud — In this course, you'll develop skills in the five key stages of visualizing data in the cloud: storytelling, planning, exploring data, building visualizations, and collaborating with data. Then, you'll explore UI/UX skills to wireframe interactive and impactful cloud-native visualizations. You'll also work with cloud-native data visualization tools to explore data sets, create reports, and build dashboards that drive decisions and foster collaboration.
5. Put It All Together: Prepare for a Cloud Data Analyst Job — In this course, you'll combine and apply the foundational knowledge and skills from previous courses to create a capstone project that focuses on the full data lifecycle. You'll also practice using cloud-based tools to acquire, store, process, analyze, visualize, and communicate data insights effectively. Then, you'll update your resume and practice interview techniques to prepare to apply and interview for jobs.

Course 2 content

Each course in this certificate program is broken into modules. You can complete courses at your own pace, but the breakdowns are designed to help you finish the entire Google Cloud Data Analytics Certificate in about 3-6 months if you complete about 1-2 modules per week.

What's to come? Here's a quick overview of the skills you'll learn in each module of this course.

Module 1: Data storage and extraction

In this module, you'll discover how data is stored and moved across systems. You'll also learn about types of data, their storage structures, and how data can be extracted, including methods of moving data between systems like APIs and connectors.

Module 2: Key components of data organization

In this module, you'll explore the key components of data governance, normalized and star schemas, data catalogs, and the data lakehouse architecture. You'll also learn how these concepts relate to modern data management, and how they can be applied in practice.

Module 3: Steps to find data

In this module, you'll learn more about how to find, trace, access, and store data with Google's cloud data storage and management tools, including BigQuery and Dataplex.

Module 4: Techniques to access data

In this module, you'll learn how to interact with data tables, and use optimization techniques to improve queries. You'll also learn how to use a cloud-based data lake or data warehouse to connect to data sources, and how to answer business questions using the data.

What to expect

Each course offers many types of learning opportunities:

- **Videos** led by Google instructors teach new concepts, introduce the use of relevant tools, offer career support, and provide inspirational personal stories.
- **Readings** build on the topics discussed in the videos, introduce related concepts, share useful resources, and describe case studies.
- **Activities** and **labs** give you hands-on practice in applying the skills you're learning and allow you to assess your own work by comparing it to a completed example.
- **Glossaries** provide a list of key terms for you to review to prepare for quizzes.
- **Practice quizzes** allow you to check your understanding of key concepts, and provide you with valuable feedback.

- **Graded quizzes** allow you to demonstrate your understanding of the main concepts of a course. You must score 80% or higher on each graded quiz to obtain a certificate. You can take a graded quiz multiple times to achieve a passing score.

Note: Some learning item types may not be included in every course.

Tips for success

- It's strongly recommended that you go through the items in each lesson in the order they appear because new information and concepts build on previous knowledge.
- Participate in all learning opportunities to gain as much knowledge and experience as possible.
- If something is confusing, don't hesitate to replay a video, review a reading, or repeat an activity.
- When you encounter useful links in this course, bookmark them so you can refer to the information later for study or review.